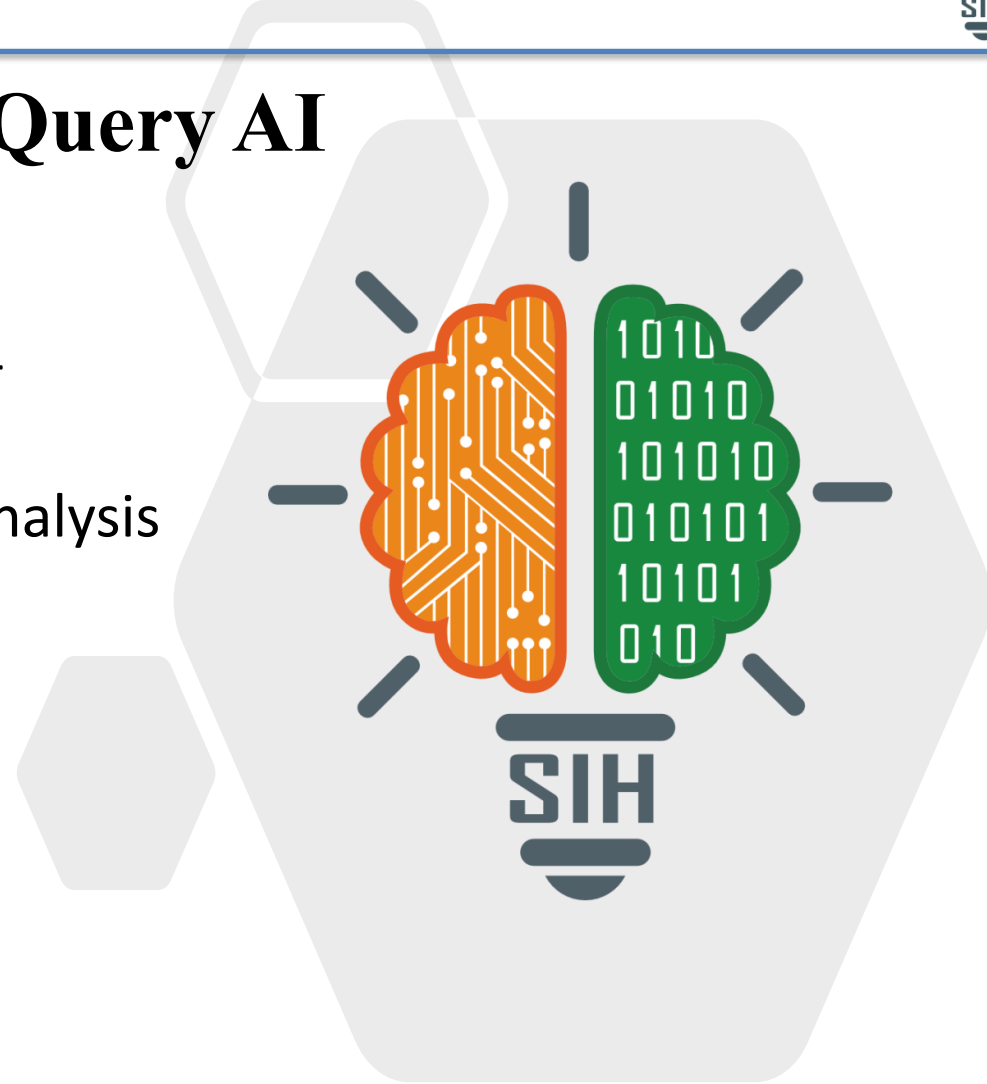




SatQuery AI

- **Problem Statement ID – SIH26167**
- **Problem Statement Title – SatQuery AI-**
An Interactive Vision-Language Assistant
for Multimodal Remote Sensing Image Analysis
through Text Queries
- **Theme – Space Technology**
- **PS Category – Software**
- **Team ID –**
- **Team Name – Team Plotvortex**



SatQuery AI – Agentic Vision-Language Assistant for Satellite Image Analysis

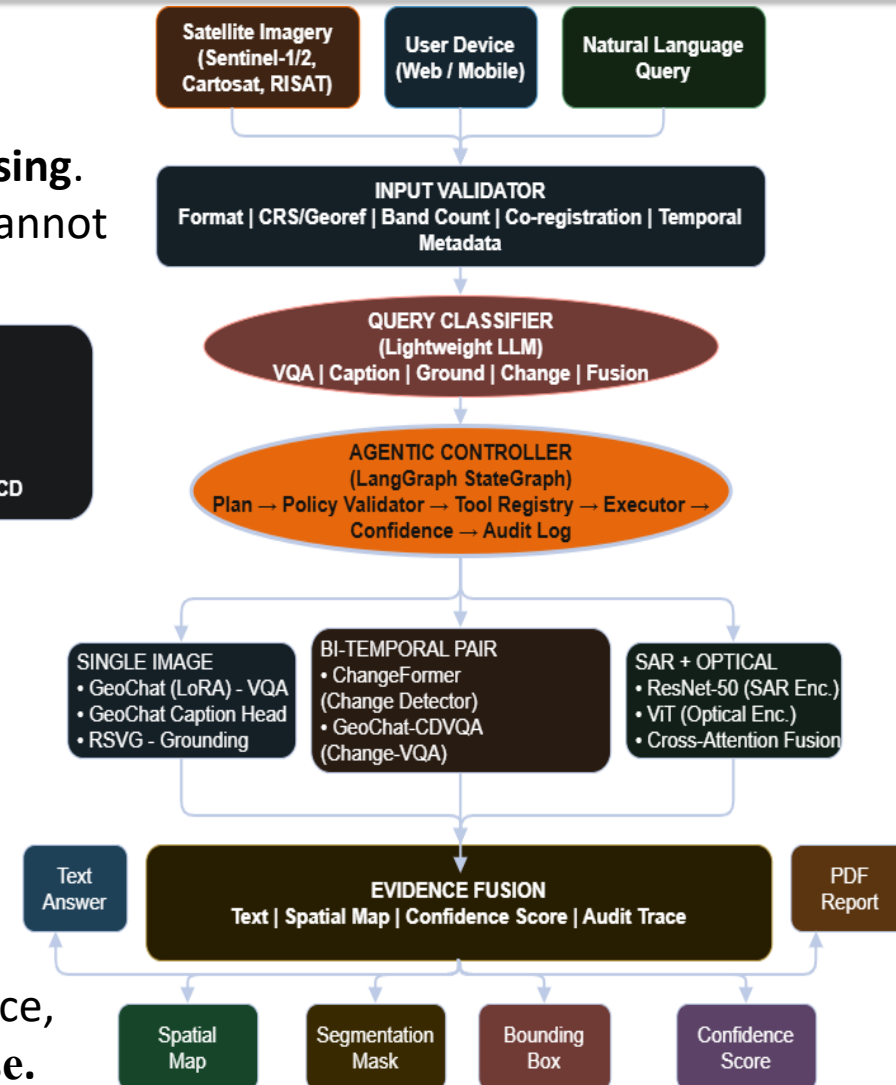
❖ Problem -

- **Extracting insights** (VQA, Change detection, SAR analysis) requires **Geographic Information System expertise, Specialized software, and hours of Manual processing.**
- **Critical users** — NDRF officers, forest departments, farmers, urban planners — cannot access this data during **time-sensitive decisions.**
- **No existing system** lets a user simply ASK a question about a satellite image and get a **grounded, visual answer instantly.**

❖ Proposed Solution -

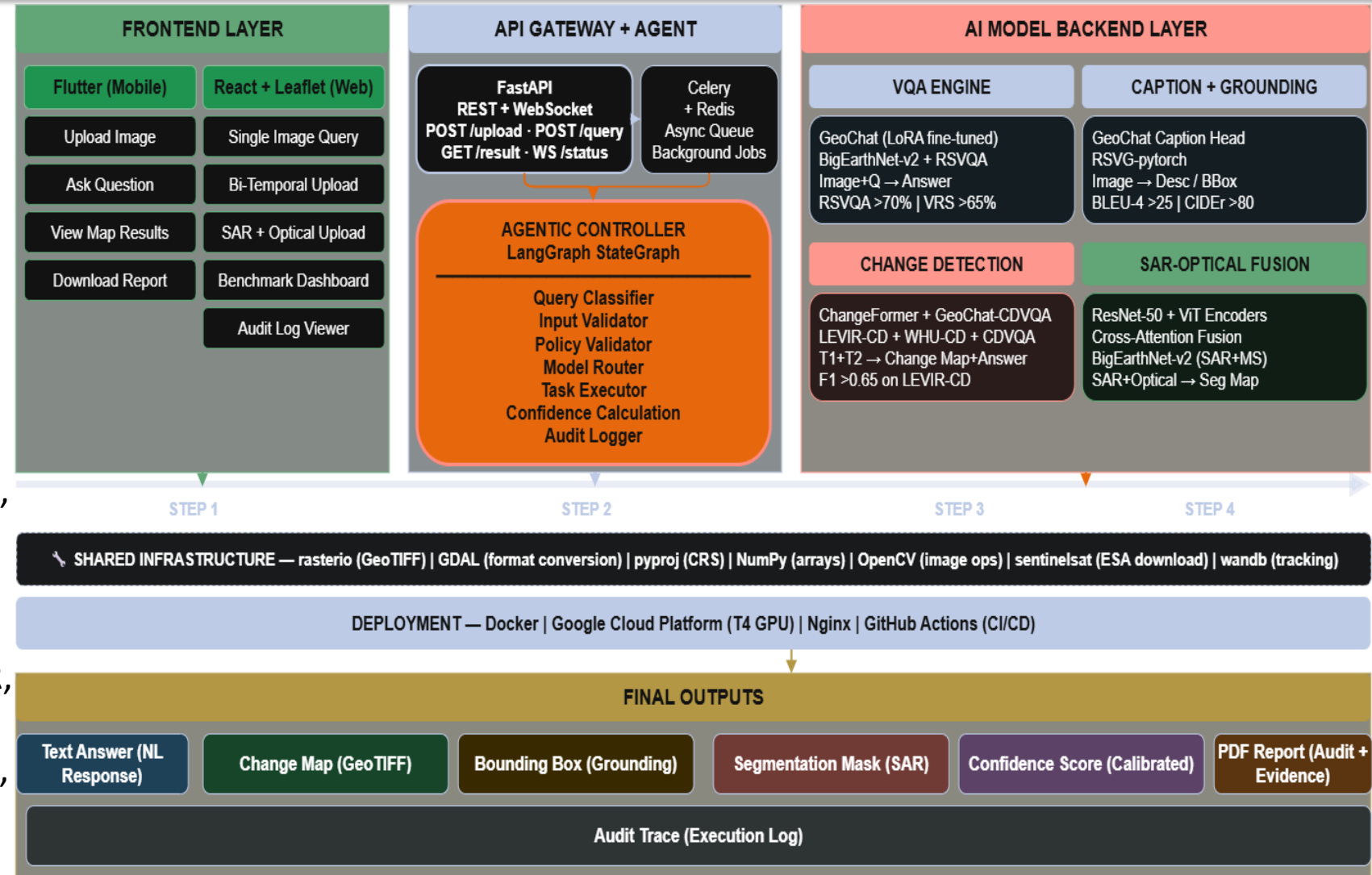
- **Agentic Vision-Language Platform** — evidence-backed answers, with **zero GIS expertise** required for any user query.
- **Multi-Task Remote Sensing** — Unifies **Scene Captioning, Visual Grounding, Bi-Temporal Change Detection, SAR–Optical Fusion, Visual Question Answering.**
- **Agentic Controller** — Classifies query intent, selecting correct **specialist model**, execution, returns a **fully auditable execution trace** — nothing **unvalidated.**
- **Evidence-Grounded Confidence Engine** — Multi-factor **confidence scoring** instead of a hallucinated LLM guess.
- **Deployment-Ready System** — GeoTIFF-native ingestion, interactive map interface, downloadable reports, containerized with Docker — **for real ISRO operational use.**

TRAINING DATA
• BigEarthNet-v2
• VRSBench
• RSVQA (LR/HR)
• CDVQA
• LEVIR-CD / WHU-CD



❖ Technology Stack -

- **Frontend** - React.js, Leaflet.js, Flutter, Tailwind CSS, Axios, React Query, Chart.js
- **Backend & API** - FastAPI, Redis, Celery, Uvicorn, Pydantic, REST & WebSocket API
- **Agentic Controller** - LangChain, Langgraph, StateGraph, Policy Validator, Tool Registry, asyncio
- **AI Models** - GeoChat, ChangeFormer, RSVG-pytorch, RemoteCLIP, ResNet-50, ViT, PEFT, LoRA, HuggingFace Transformers, accelerate, timm, einops
- **Satellite Data** - rasterio, GDAL, pyproj, NumPy, OpenCV, sentinelsat, scipy, Pillow
- **Training Datasets** - VRSBench, RSVQA-LR, RSVQA-HR, CDVQA, LEVIR-CD, WHU-CD
- **Deployment** - Docker, GCP, Nginx, GitHub, Actions, wandb, evaluate, pycocoevalcap, scikit-learn



❖ Feasibility (Technical)

- **Open-source base models** – GeoChat, ChangeFormer, RSVG already exist.
- **LoRA fine-tuning** – 7B model trained on single GPU, not clusters.
- **BigEarthNet dataset** – 590K pre-labeled images, free download.
- **Modular architecture** – VQA, Change Detection, SAR-Fusion built independently.
- **Free satellite data** – ESA, ISRO Bhuvan, NASA all no-cost sources.

❖ Viability

- **Direct ISRO alignment** – Solves their exact stated problem
- **50+ Indian satellites** – Massive unused data pool to unlock
- **Zero licensing cost** – 100% open-source stack, no paid APIs
- **Phased scalability** – NDRF → Agriculture → Full ISRO deployment
- **Realistic benchmarks** – 65-70% accuracy target, proven achievable

❖ Challenges

- **SAR complexity** – Radar noise differs from optical images
- **Compute limits** – Multi-dataset fine-tuning is GPU-heavy
- **Unknown test data** – ISRO's hidden Cartosat/RISAT set untestable
- **Agent hallucination** – LLM may fabricate numeric answers
- **Fragile setup** – GDAL/rasterio break easily if installed wrong

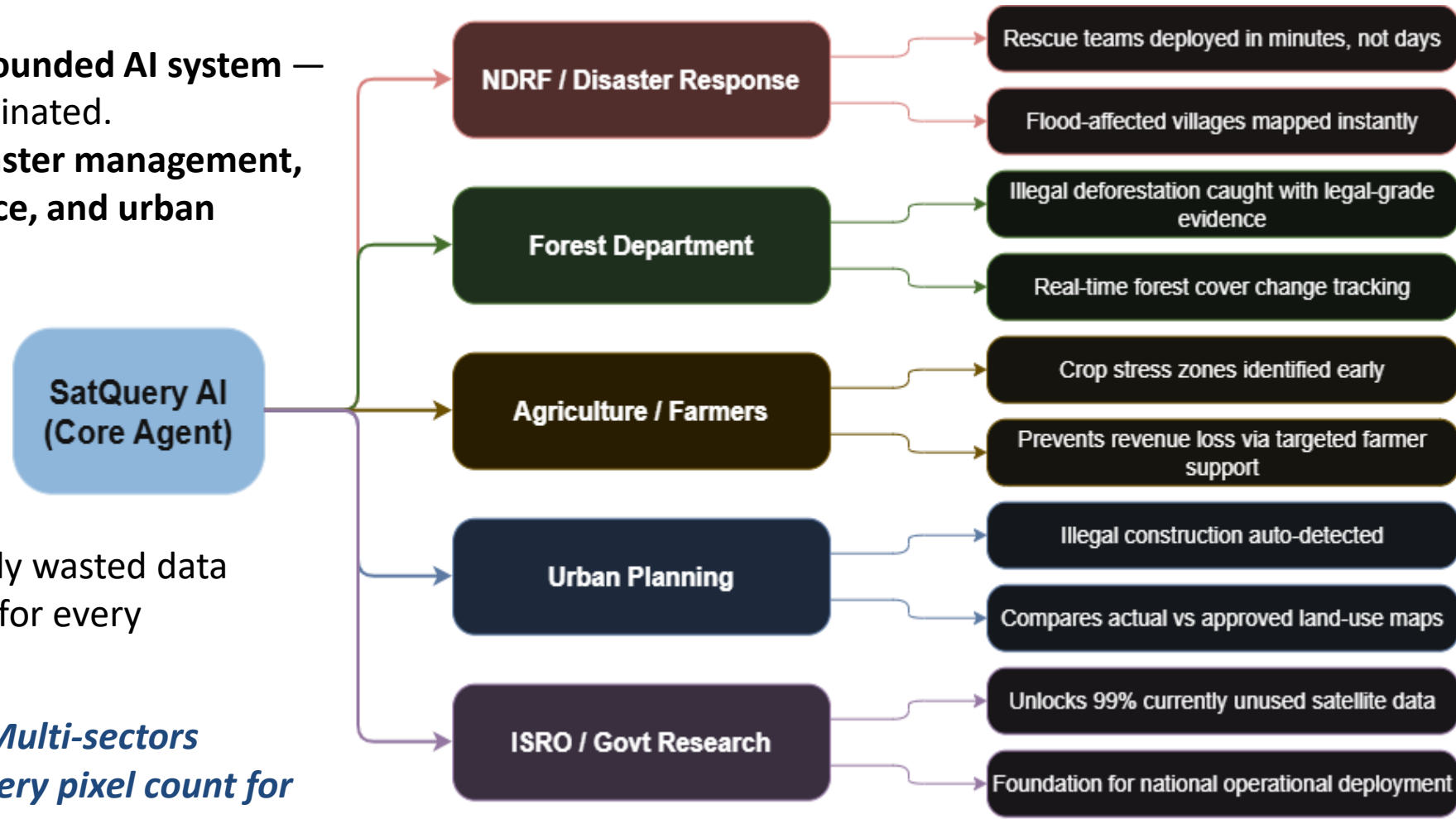
❖ STRATEGY

- **Locked agent design** – LLM only classifies; models compute real numbers
- **Sensor-agnostic pipeline** – Normalizes data for unseen satellite types
- **Pre-built before hackathon** – 36 hrs used only for integration + polish
- **Live benchmark dashboard** – Real VRSBench/RSVQA/CDVQA scores shown
- **Conda-first install rule** – GDAL via conda, prevents environment crashes

❖ Strategic Impact-

- Offers a **fully auditable, evidence-grounded AI system** — every answer traceable, nothing hallucinated.
- Strengthens India's autonomy in **disaster management, precision agriculture, forest governance, and urban oversight** — all from one platform.
- Directly **advances Atmanirbhar Bharat** in space-tech by building India's first indigenous agentic satellite-AI system deployable at national scale.
- Converts **50+ ISRO satellites'** currently wasted data into **actionable, real-time intelligence** for every government department and citizen.

"One question. Any satellite image. Multi-sectors transformed — SatQuery AI makes every pixel count for India."



❖ Research Papers:

- **GeoChat** — Grounded Large Vision-Language Model for Remote Sensing:
<https://arxiv.org/abs/2311.15826>
- **BigEarthNet-v2** — A Large-Scale Benchmark for RS Image Understanding:
<https://arxiv.org/abs/2603.29630>
- **RSVQA** — Visual Question Answering for Remote Sensing Data:
<https://arxiv.org/abs/2003.07333>
- **RemoteCLIP** — A Vision Language Foundation Model for Remote Sensing:
<https://arxiv.org/abs/2306.11029>

❖ Problem Validation:

- **ISRO Annual Report 2023**—Data Utilization Gap:
https://www.isro.gov.in/AnnualReport_2023.html
- **NDRF Disaster Response Report 2024**:
<https://ndma.gov.in/Resources/reports>

Feature	SatQuery AI	Generic BLIP-2/LLaVA	QGIS + GIS Expert	Google Earth Engine
RS-Adapted VLM	✓	✗	✗	✗
Natural Language Query	✓	⚠	✗	✗
Single Image VQA	✓	⚠	✗	✗
Bi-Temporal Change Detection	✓	✗	⚠	⚠
SAR + Optical Fusion	✓	✗	⚠	⚠
Agentic Orchestration	✓	✗	✗	✗
Auditable Execution Trace	✓	✗	✗	✗
GeoTIFF Native Support	✓	✗	✓	✓
Zero GIS Expertise Needed	✓	⚠	✗	✗
Calibrated Confidence Score	✓	✗	✗	✗
Free / Open Source	✓	✓	⚠	⚠
Downloadable PDF Report	✓	✗	⚠	✗
✓ Full Support ⚠ Partial / Manual ✗ Not Supported				